Unit 3 Review Sheet

Math 9E · Measurement · Every formula, theorem and rule from Lessons 3.1–3.4, plus the trigonometry used in class.

3.1 Similar Triangles & Theorems

RULE / THEOREM	FORMULA OR STATEMENT	WHEN TO USE IT
Similar triangles	Equal corresponding angles **and** proportional sides. Write $\Delta ABC \sim \Delta DEF$ ($\sim$ = similar, $\cong$ = congruent)	Any "find the missing side" with two triangles of the same shape
Angle sum	$\angle A + \angle B + \angle C = 180°$	Find the third angle; two equal angles ⇒ third is equal
Parallel lines	Corresponding angles equal; alternate angles equal	Proving AA when a line is parallel to a side
AA test	Two pairs of equal angles ⇒ similar	Most common proof (right angles, shared angles, vertical angles)
SAS test	Two sides in the same ratio **and** the angle between them equal	Two sides + included angle given
SSS test	$a/a' = b/b' = c/c'$	Practice 3.1 #1: compare all three ratios
Thales' theorem	If DE ∥ BC: **$AD/DB = AE/EC$** and **$AD/AB = AE/AC = DE/BC$**	Line inside a triangle parallel to a side; cone cross-sections
Converse of Thales	If $AD/DB = AE/EC$, then DE ∥ BC	"Is DE parallel to BC?"
Hourglass	AB ∥ ED and AD, BE cross at C ⇒ $\Delta ABC \sim \Delta DEC$ ⇒ $AC/DC = BC/EC = AB/DE$	Practice 3.1 #6
Shadow problems	$\text{height}_1 / \text{shadow}_1 = \text{height}_2 / \text{shadow}_2$	Sun rays are parallel ⇒ similar triangles
Scale factor k	Lengths × k · Perimeter × k · **Area × k^2** · **Volume × k^3**	Similar shapes, cones, pyramids, maps, models
Altitude on hypotenuse	Right angle at C, CD ⊥ AB: **$CD^2 = AD \cdot DB$**, $AC^2 = AD \cdot AB$, $BC^2 = BD \cdot AB$	Right triangle with an altitude drawn to the hypotenuse

Pythagoras	$a^2 + b^2 = c^2$ (c = hypotenuse)	Slant height, missing side of a right triangle

★ Trigonometry

RULE	FORMULA	NOTE
SOH CAH TOA	$\sin\theta = \text{opp/hyp} \cdot \cos\theta = \text{adj/hyp} \cdot \tan\theta = \text{opp/adj}$	Right triangles only
Special values	$\sin 30° = ½ \cdot \cos 30° = \sqrt{3}/2 \cdot \sin 60° = \sqrt{3}/2 \cdot \cos 60° = ½ \cdot \tan 45° = 1$	Equilateral triangles use 30° and 60°
Circle around an equilateral triangle	**$R = a/\sqrt{3}$** (from $\cos 30° = (a/2)/R$) $\cdot\ a = R\sqrt{3}$	Inscribed triangular prism in a cylinder
Law of sines	$a / \sin A = 2R$	Quick check for R
Equilateral triangle area	$A = (\sqrt{3}/4)a^2 \cdot \text{height} = (\sqrt{3}/2)a$	Triangular prism volume
Square in a circle	$\text{side} = R\sqrt{2} \cdot \text{area} = 2R^2$	Square prism inside a cylinder

3.2 Unit Conversions

RULE	FORMULA OR VALUE	NOTE
Conversion method	Starting value × (units converting **to**) / (units converting **from**)	Units must cancel
Metric prefixes	$T\ 10^{12} \cdot G\ 10^{9} \cdot M\ 10^{6} \cdot k\ 10^{3} \cdot h\ 10^{2} \cdot da\ 10 \cdot d\ 10^{-1} \cdot c\ 10^{-2} \cdot m\ 10^{-3} \cdot \mu\ 10^{-6} \cdot n\ 10^{-9} \cdot p\ 10^{-12} \cdot f\ 10^{-15}$	Go through the base unit (m)
Imperial lengths	1 ft = 12 in · 1 yd = 3 ft · 1 mi = 5280 ft = 1760 yd	
SI ↔ Imperial	1 in = 2.54 cm · 1 m = 3.28 ft = 39.37 in = 1.094 yd · 1 km = 0.621 mi · 1 mi = 1.609 km	
Land area	$1\ \text{acre} = 4840\ \text{yd}^2 \cdot 1\ \text{mi}^2 = 640\ \text{acres} \cdot 1\ \text{ha} = 10\,000\ \text{m}^2 = 2.47\ \text{acres} \cdot 1\ \text{km}^2 = 100\ \text{ha}$	
Area units	Square the factor: $1\ \text{m}^2 = 100^2\ \text{cm}^2 = 10\,000\ \text{cm}^2 \cdot 1\ \text{ft}^2 = 30.48^2\ \text{cm}^2$	Most common mistake
Volume units	Cube the factor: $1\ \text{m}^3 = 100^3\ \text{cm}^3 = 1\,000\,000\ \text{cm}^3$	

Speed	km/h ÷ 3.6 = m/s	× 1000 ÷ 3600
Medicine dose	Amount to give = (ordered ÷ on hand) × quantity · e.g. (90 mg ÷ 120 mg) × 5 mL	Convert g ↔ mg first (1 g = 1000 mg)

3.2 Surface Area

SHAPE	SURFACE AREA	NOTE
Rectangle · Triangle · Circle	$A = lw$ · $A = bh/2$ · $A = \pi r^2$, $C = 2\pi r$	Build blocks for every solid
Rectangular prism	$SA = 2(lw + lh + wh)$	
Cylinder	$SA = 2\pi r^2 + 2\pi rh$	Open top: one circle only
Cone	$SA = \pi rs + \pi r^2$ (lateral πrs)	**$s = \sqrt{(h^2 + r^2)}$** is the slant, not the height
Sphere	$SA = 4\pi r^2$	
Hemisphere	Curved $2\pi r^2$ · solid total $3\pi r^2$	Bowl vs solid: is the flat face included?
Square pyramid	$SA = b^2 + 4(\frac{1}{2} \cdot b \cdot s) = b^2 + 2bs$	**$s = \sqrt{(h^2 + (b/2)^2)}$**
Composite shapes	Add the outside faces only; remove faces that touch	Silo: walls + roof, no base
Hole through a solid	Subtract the hole openings, add the inside wall	SA can go **up** after drilling

3.3 Volume & Capacity

SHAPE / RULE	FORMULA	NOTE
Any prism	V = (area of base) × height = Bh	Box: $V = lwh$
Cylinder	$V = \pi r^2 h$	
Pyramid	$V = \frac{1}{3}Bh = lwh/3$	⅓ of the prism with the same base and height
Cone	$V = \frac{1}{3}\pi r^2 h$	⅓ of the cylinder
Sphere · Hemisphere	$V = (4/3)\pi r^3$ · $V = (2/3)\pi r^3$	

Capacity	$1\ cm^3 = 1\ mL \cdot 1000\ cm^3 = 1\ L \cdot 1\ m^3 = 1000\ L = 1\ kL \cdot 1\ US\ gal = 3.785\ L$	Volume = space; capacity = what it holds
Cavalieri 2D	Same height + same width at every level ⇒ same area	
Cavalieri 3D	Same height + same cross-section **area** at every level ⇒ same volume	Shape can differ (Carl vs Hiva)
Sphere proof	Hemisphere = cylinder − cone = $\pi r^3 - \frac{1}{3}\pi r^3 = \frac{2}{3}\pi r^3$	Slice areas: $\pi(r^2 - h^2)$ on both
Pyramid proof	A cube splits into 3 equal pyramids ⇒ $V = s^3/3$	"Challenge before next class"
Cut pyramid or cone (frustum)	$V = V_{big} - V_{small}$, with $V_{small} = V_{big} \times k^3$	k = (cut distance from apex) ÷ (full height)
Water in a cone (point down)	$V_{water} = V_{tank} \times (\text{depth/height})^3$	Half full is not half the depth
Composite / removed parts	Add parts, subtract holes	Sloped pool: trapezoid side × width

3.3 Key Formulas for Solving Volume & Area Problems

Rearranged formulas: use these when the question gives the volume or area and asks for r, h or s.

FIND	FORMULA	TYPICAL QUESTION
Cylinder r or h	$h = V / (\pi r^2)$ · $r = \sqrt{(V / (\pi h))}$ · from SA: $h = (SA - 2\pi r^2) / (2\pi r)$	"SA = 56π, find h" (Practice 3.2 #6)
Cone r or h from V	$h = 3V / (\pi r^2)$ · **$r = \sqrt{(3V / (\pi h))}$**	"V = 205 yd³, find r" (Practice 3.3 #3)
Cone r or s from lateral SA	**$r = L / (\pi s)$** · $s = L / (\pi r)$	"Curved SA = 177 cm², find r" (Practice 3.2 #7)
Cone h, r or s	$s = \sqrt{(h^2 + r^2)}$ · $h = \sqrt{(s^2 - r^2)}$ · $r = \sqrt{(s^2 - h^2)}$	Switching between height and slant
Sphere r	From V: **$r = \sqrt[3]{(3V / (4\pi))}$** · from SA: $r = \sqrt{(SA / (4\pi))}$	"SA = 196π, find r"
Hemisphere r	From V: $r = \sqrt[3]{(3V / (2\pi))}$ · from solid SA: $r = \sqrt{(SA / (3\pi))}$	

Pyramid h, b or s	**$h = 3V / B$** · square base: $h = 3V / b^2$, $b = \sqrt{(3V / h)}$ · $s = \sqrt{(h^2 + (b/2)^2)}$	"V = 111 yd³, base 6, find s" (Practice 3.3 #5)
Prism h	$h = V / B$	Fish tank height for a given capacity
Scale factor of similar solids	$k = \sqrt[3]{(V_2 / V_1)}$ · $k = \sqrt{(A_2 / A_1)}$	"Volumes 54π and 250π, ratio of radii?"
Depth for a fraction f of a cone tank	**$d = H \cdot \sqrt[3]{f}$**	"How deep is half full?" ⇒ d ≈ 0.794 H
Water in a cone tank	$V_{water} = V_{tank} \times (d / H)^3$	"Water level 1 m below the top" (Practice 3.3 #4b)
Cut pyramid or cone (remaining piece)	**$V = V_{big} \times (1 - (d / H)^3)$**, d = cut distance from apex	Practice 3.3 #9
Removed part	$V_{left} = V_{solid} - V_{hole}$ ⇒ solve for the hole's r or side	Quarter cylinder: ¼πr²L = $V_{removed}$ (Practice 3.3 #8)
Melting / recasting	Number of pieces = $V_{original} \div V_{one\ piece}$	Sphere melted into cones
Filling time	Time = capacity ÷ rate (same units: L and L/min)	Tank or pool fill time
Sloped pool	$V = (d_1 + d_2)/2 \times$ length × width	Practice 3.3 #10

⚡ Shortcut Formulas for Speed

SHORTCUT	FORMULA OR FACT	WHY IT SAVES TIME
Cylinder : sphere : cone	Same r, height 2r ⇒ volumes **3 : 2 : 1**	Cone = ⅓ cylinder · sphere = ⅔ cylinder
Hemisphere vs cylinder	Same r, h = r ⇒ hemisphere = ⅔ cylinder, cone = ⅓ cylinder	Cavalieri result, no calculation needed
Factored SA formulas	Cylinder $2\pi r(r + h)$ · cone $\pi r(r + s)$ · square pyramid $b(b + 2s)$	Fewer steps on the calculator
Silo (cylinder + dome, no base)	$SA = 2\pi rh + 2\pi r^2$	Same expression as a closed cylinder
Pythagorean triples	3-4-5 · 6-8-10 · 5-12-13 · 8-15-17 · 7-24-25 · 9-12-15	Slant height instantly

Doubling rules	Lengths × 2 ⇒ area × 4, volume × 8 · cylinder: r × 2 ⇒ V × 4, h × 2 ⇒ V × 2	"What happens if..." questions
Half-depth cone tank	Filled to half the depth ⇒ holds only **1/8** of the volume	$(½)^3 = ⅛$
Shape inside a cube of side s	Sphere = π/6 ≈ 52.4% · cylinder = π/4 ≈ 78.5% · pyramid = ⅓	Wasted-space questions
Prism inside a cylinder	Triangle ≈ 41.3% · square ≈ 63.7% · hexagon ≈ 82.7% of the circle	Waste = 100% − used
Equilateral triangle	Area ≈ $0.433a^2$ · R = a/√3 ≈ 0.577a	Triangular prisms
Capacity	$cm^3 ÷ 1000 = L$ · $m^3 × 1000 = L$ · L ÷ 3.785 = gal	One step instead of three
Area conversions	$m^2 → cm^2$: × 10 000 · $km^2 → m^2$: × 1 000 000 · km^2 → ha: × 100	
Speed	km/h ÷ 3.6 = m/s · m/s × 3.6 = km/h	
Tax	12% tax ⇒ × 1.12 · 13% ⇒ × 1.13	One multiplication
Ratio to k	Area ratio ⇒ take √ · volume ratio ⇒ take $\sqrt[3]{}$	Similar-solid questions
Quick estimate	Use π ≈ 3 to check that an answer is reasonable	Catches calculator slips
Keep π	Work in terms of π and multiply by π only at the end	Exact answers, fewer rounding errors

3.4 Scale, Scale Diagrams & Isometric Drawings

RULE	FORMULA OR STATEMENT	NOTE
Scale	Scale = model : actual (same units!)	3.5 cm : 4.2 m = 3.5 : 420 = 1 : 120
Reduction	Model smaller than reality · 1 : bigger number (e.g. 1 : 50)	Maps, floor plans
Enlargement	Model bigger than reality · bigger number : 1 (e.g. 8000 : 1)	Microscope images
Scale factor	Bigger number ÷ smaller number	Not the same as the scale

Using the factor	**Multiply** to find the bigger value · **Divide** to find the smaller value	
Area and volume on a scale	Area × n^2 · Volume × n^3	Map area, model tank capacity
Grid scale	"4 cm represents 1 m" ⇒ divide plan lengths by 4	Practice 3.4 #8
Isometric drawing	3-D shape on a dot grid; vertical lines stay vertical, others at 30°	Draw from top, front (left), right views
Orthographic views	Top · front · side: show the tallest stack in each row/column	Hidden cubes do not appear

! Common Traps

TRAP	WHAT TO DO
Diameter given	Halve it before using r
Height vs slant height	Cones and pyramids: SA needs s, volume needs h. Use Pythagoras to switch
Mixed units	Convert everything to one unit before calculating
Area or volume conversions	Square or cube the length factor
Similar shapes	Lengths k, areas k^2, volumes k^3
Buckets, cans, tiles	Always round **up** to a whole number
Tax	Total × 1.12 for 12% tax
Base included?	Read whether the bottom, top or flat face is part of the surface
Matching vertices	Write similar triangles in matching order before writing ratios
Answer in terms of π	Keep π as a symbol and do not round